

PIXELORA

Complete Official User Guide & Technical Manual

The definitive architectural and operational reference for Pixelora image processing suite, background cutout tools, and client-side raster engine.

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CHAPTER 01

Getting Started

Comprehensive overview of Pixelora, rapid 5-step workflow from import to final export.

Pixelora is an advanced browser-based photo editor and graphic design studio. It blends consumer accessibility with industrial-grade digital image processing, featuring a non-destructive layer stack, 64 mathematical filters, Lightroom-style adjustments, and advanced background cutout tools.

What is Pixelora? Core Purpose

Functionality: Delivers a full-featured workstation for pixel-level editing, non-destructive layer compositing, tone curves, background cutout, and commercial graphic design.

When to Use: Whenever preparing e-commerce catalog photos, enhancing portraits, crafting marketing graphics, or applying advanced mathematical convolutions.

Step-by-Step Procedure:

1. Open Pixelora directly in your modern web browser without plugin installations.
2. Choose "New Project" or drag and drop an existing photograph onto the canvas.
3. Leverage the vertical toolbox and properties inspector to manipulate layers and adjustments.
4. Export your final artwork in PNG, JPG, or WEBP at pristine resolution.

5-Step Quick Start Workflow

Functionality: Outlines the logical sequence: Ingestion → Geometry/Crop → Tone Adjustments → Effects & Masks → Final Export.

When to Use: Follow this sequence for every standard editing session to maintain peak visual fidelity and layer hygiene.

Step-by-Step Procedure:

1. Step 1 (Ingestion): Click "Open Image" or drop an image file onto the startup area.
2. Step 2 (Geometry): Define canvas bounds or press C for the Crop tool to frame the subject.
3. Step 3 (Color & Tone): Navigate to Adjustments (A) to calibrate exposure, curves, and HSL mixer.
4. Step 4 (Effects & Overlays): Apply stylistic presets, isolate subjects with 1-click cutout, or add typography.
5. Step 5 (Export): Press Ctrl+E to launch the Export modal, choosing PNG for transparency or JPG/WEBP for compact web distribution.

🔖 Pro Tip:

- Use Ctrl+Z to undo any step, and Ctrl+Y (or Shift+Ctrl+Z) to redo.
- You can safely reset individual adjustment panels without affecting other active layers.

CHAPTER 02

Understanding the Workspace

Exhaustive tour of the interface: Top bar, vertical toolbox, viewport canvas, and properties inspector.

Pixelora's interface follows desktop creative workstation paradigms, organizing controls into functional clusters with responsive collapsing, dual theme engine, and bidirectional RTL/LTR layout.

Header & Topbar Anatomy

Functionality: Provides direct project title editing, project serialization to local storage, undo/redo triggers, and instant export initialization.

When to Use: For project-level lifecycle commands, tracking undo history state, and launching modals.

Step-by-Step Procedure:

1. Click the project name input to rename your document inline.
2. Use the curved arrow buttons for undo and redo with instant visual feedback.
3. Click the primary purple "Export" button to configure resolution, scale, and format.

Main Vertical Toolbox

Functionality: Clusters tools into Core (Select, Transform, Crop, Hand, Zoom), Create (Brush, Pen, Eraser, Text, Shapes), Enhance (Remove Object, Filters, Adjustments), and Composite (Remove BG, Background Library).

When to Use: To toggle the active interaction mode across canvas pointer events.

Step-by-Step Procedure:

1. Click any tool icon or press its designated keyboard shortcut (e.g., V, B, P, E, T).
2. Activating a tool automatically synchronizes the right properties panel to display its parameters.

Canvas Viewport & Stage

Functionality: Supports pan & zoom from 10% to 800%, transparency checkerboard rendering, and transform bounding box overlays.

When to Use: For visually manipulating elements and observing real-time composite changes.

Step-by-Step Procedure:

1. Scroll the mouse wheel to zoom in and out centered on the cursor position.
2. Hold the Spacebar and drag with the left mouse button to pan smoothly across the canvas.
3. Click directly on any layer object to select it and engage boundary transform handles.

Properties & Inspector Panel

Functionality: Adapts contextually to current selections, providing fine-grained numerical and visual controls.

When to Use: To fine-tune color balances, reorder layers, tweak typography, and adjust brush hardness.

Step-by-Step Procedure:

1. Switch tabs between Layers, Adjustments, Filters, Backgrounds, and Typography.
2. Drag sliders or type exact values into the numeric inputs for maximum precision.

CHAPTER 03

Files & Projects Management

Creating new documents, local project persistence, duplicate branching, and standard social presets.

Pixelora includes a robust local document system. You can spin up blank canvases tailored to social presets, or ingest single/multiple image files that automatically organize into discreet layers.

New Project Dialog & Preset Dimensions

Functionality: Configures canvas width, height, initial background fill (transparent, white, dark), and project nomenclature.

When to Use: When constructing banners, product hero shots, or social posts from an empty stage.

Step-by-Step Procedure:

1. Press Ctrl+N or select "New Project" from the top navigation bar.

2. Pick a preset: Instagram Square (1080x1080), Story (1080x1920), Banner (1200x630), or E-Commerce (1200x1200).
3. Alternatively enter custom width and height and click "Create Canvas".

Multi-Image Ingestion & Layer Generation

Functionality: Parses selected images into individual layer objects preserving original pixel resolution.

When to Use: When combining product variations, composing collages, or staging multi-layer assets.

Step-by-Step Procedure:

1. Select multiple images in your operating system file manager and drop them onto the home landing zone.
2. Pixelora will instantiate a canvas scaled to fit and load each file as a separate layer.

Saved Projects Library & Duplication

Functionality: Displays project thumbnail cards with timestamps and layer counts, providing 1-click session restoration.

When to Use: To revisit previous work or fork an existing design template for a new variant.

Step-by-Step Procedure:

1. Click the "Projects" tab in the main navigation menu.
2. Click any project card to immediately mount its complete layer state in the editor.
3. Click "Duplicate" to spawn an independent copy for testing alternatives.

CHAPTER 04

Basic Image & Geometry Tools

Crop, Document Image Size, Canvas Size, Advanced Transform, Rotation, and Flipping.

Geometric operations define image boundaries and spatial resolution. Pixelora strictly distinguishes between Image Resampling (recalculating pixel matrices), Canvas Sizing (extending or cropping the frame boundary), and Viewport Zooming (magnification without altering pixels).

Critical Distinction: Image Size vs Canvas Size vs Zoom

Functionality: Explains that Image Size recomputes bitmap resolution, Canvas Size adds or crops frame padding around existing layers, and Zoom changes only viewport scale.

When to Use: Use Image Size for web optimization or print scaling, Canvas Size for padding/margins, and Zoom to inspect fine pixel details.

Step-by-Step Procedure:

1. For Image Resampling: Open Image Size modal, set dimensions, and pick an algorithm (Bilinear, Bicubic, Lanczos).
2. For Canvas Sizing: Open Canvas Size modal, pick dimension delta, anchor point on 9-point grid, and canvas background fill.
3. For Zoom: Tap Z, roll mouse wheel, or drag the bottom percentage slider.

Interactive Crop Tool (C)

Functionality: Provides an interactive rectangular viewport overlay with Rule-of-Thirds guides and aspect ratio locks.

When to Use: To reframe compositions, remove edge distractions, or conform assets to social media feeds.

Step-by-Step Procedure:

- 1. Press C to summon the crop overlay box.
- 2. Drag corner handles inward or select an aspect preset (1:1, 4:5, 16:9).
- 3. Double click inside the boundary or press Enter to apply crop parameters.

Parameter	Type	Default	Description
Aspect Ratio	Preset / Free	Free	Locks the ratio between width and height during drag transformations.
Crop Canvas vs Layer	Mode	Canvas	Specifies whether to trim the entire stage or solely the selected image layer.

Advanced Transform Tool (Ctrl+T)

Functionality: Renders an interactive bounding box with 8 scale handles, a top rotation handle, skew shears, and numeric inspector sync.

When to Use: When fitting graphic overlays, aligning product mockups, shearing isometric vectors, or rotating angles.

Step-by-Step Procedure:

- 1. Press Ctrl+T to toggle advanced transformation mode on the selected layer.
- 2. Drag bounding corners to scale; hover outside corners to rotate.

3. Use the inspector panel to type precise angles, skew degrees, or apply 1-click alignments.

CHAPTER 05

Drawing, Vectors & Creative Tools

6-style Brush Tool, Bézier Pen Tool & vector paths, dual-mode Eraser, and 70+ Shape library.

Pixelora integrates procedural raster brush engines alongside editable Bézier vector paths and a library of over 70 geometric, callout, and decorative shapes.

Procedural Brush Tool (B)

Functionality: Renders smooth raster strokes simulating organic media: Soft, Hard, Neon Glow, Calligraphy/Marker, Airbrush/Spray, and Watercolor.

When to Use: For hand-drawn illustrations, product highlighting, digital signatures, and organic vignette touchups.

Step-by-Step Procedure:

- 1. Press B to activate the brush tool.
- 2. Select your desired brush engine from the inspector (e.g., Neon or Watercolor).
- 3. Configure stroke radius, opacity, hardness, and color via the color palette or eyedropper.
- 4. Paint directly onto the canvas with smooth interpolated stroke updates.

Parameter	Type	Default	Description
Brush Radius / Size	px (1 - 300)	24px	Diameter of the painted stroke in canvas pixels.
Hardness	0.0 - 1.0	0.8	Controls edge falloff softness: 0 produces ultra-diffuse airbrushing, 1 creates sharp razor edges.
Opacity	0% - 100%	100%	Alpha density of stroke pigment, allowing progressive watercolor glazing.

Pen Tool & Bézier Vector Paths (P)

Functionality: Plots anchor points and directional handles for cubic Bézier mathematics. Complete paths can be rasterized to marching-ants selection masks, stroked with vector outlines, or filled with color.

When to Use: For pinpoint-accurate product clipping paths, custom vector silhouette extraction, and non-standard geometric design.

Step-by-Step Procedure:

1. Press P to engage the vector pen mode.
2. Click to insert linear corner anchors, or click-and-drag to extend smooth tangent handles.
3. Click back onto the root anchor point to close the path loop.
4. Click "Make Selection" in the inspector to activate marching ants, or choose "Stroke" / "Fill".

Dual-Mode Eraser: Pixel vs Mask Cutout (E)

Functionality: Offers two modes: Raster Pixel Erasing for drawing layers, and Non-Destructive Alpha Mask Cutout for image layers with instant 1-click restore.

When to Use: To clean unwanted brush strokes, hand-refine edge cutouts, or feather layer transitions non-destructively.

Step-by-Step Procedure:

1. Press E to activate the eraser.
2. Calibrate brush size and edge softness.
3. Erase over unwanted pixels; on image layers, it safely creates a mask cutout that preserves raw data.

Shapes & Vector Icons Library (U)

Functionality: Inserts resolution-independent vector shapes categorized by Basic Geometry, Directional Arrows, Badges & Seals, Callouts & Speech Bubbles, and Fluid Organics.

When to Use: For creating discount price tags, product badges, promotional ribbons, framing cards, and CTA containers.

Step-by-Step Procedure:

1. Press U to open the shape library modal.
2. Pick any shape geometry (e.g., starburst, badge, speech bubble, or rounded rectangle).
3. Fine-tune fill color, stroke color, stroke weight, and corner radii in the properties panel.

CHAPTER 06

Typography & Multilingual Text System

Multilingual text engine, Arabic & Latin fonts, gradient fills, outlines, and drop shadows.

Pixelora features an advanced dual-engine typography system supporting full bidirectional Arabic (RTL) text shaping alongside Latin fonts, gradient fills, multiple strokes, and soft drop shadows.

Text Creation, Alignment & Font Selection

Functionality: Facilitates inline headline and body composition using popular web fonts (Cairo, Tajawal, Amiri, Alexandria, Inter, Montserrat, Playfair).

When to Use: For branding titles, e-commerce promotional banners, calls-to-action, and watermarks.

Step-by-Step Procedure:

1. Press T on your keyboard to instantiate a new typography layer.
2. Type your copy directly into the inspector text field with real-time canvas rendering.
3. Select font family, font size, letter spacing, line height, and text alignment.

Styling Effects: Gradients, Strokes & Shadows

Functionality: Renders linear text gradients across customizable angles, multi-pixel vector stroke envelopes, and soft Gaussian drop shadows.

When to Use: To guarantee high contrast over busy photography backgrounds and inject premium aesthetic polish.

Step-by-Step Procedure:

1. In the text properties panel, toggle "Gradient Fill" and select a color palette.
2. To outline the characters, enable "Stroke" and set pixel thickness and outline color.
3. To introduce depth, enable "Shadow" and adjust X/Y offset, blur radius, and shadow color.

CHAPTER 07

Layers, Masks & Blend Modes Architecture

Layer hierarchy, drag reordering, visibility & lock, 16 blend modes, and non-destructive masking.

Pixelora utilizes a non-destructive layer stack. Each element (image, text, shape, drawing, or effect layer) possesses its own isolated transform matrix, alpha mask, opacity, and blend mode mathematics.

Layer Tree Management, Ordering & Protection

Functionality: Renders the bottom-to-top compositional stack, enabling drag reordering, duplication, locking against accidental edits, and deletion.

When to Use: Throughout every project to organize scene depth, manage z-index ordering, and anchor background assets.

Step-by-Step Procedure:

1. Open the "Layers" tab in the right-hand panel.
2. Click and drag any layer card up or down to adjust visual stacking precedence.
3. Click the eye icon to toggle visibility on and off without deleting assets.
4. Click the padlock icon to freeze a layer against accidental canvas drag operations.

16 Blend Modes & Mathematical Compounding

Functionality: Implements standard W3C compositing algorithms: Normal, Multiply, Darken, Color Burn, Screen, Lighten, Color Dodge, Overlay, Soft Light, Hard Light, Difference, Exclusion, Hue, Saturation, Color, and Luminosity.

When to Use: To seamlessly blend light leaks, apply grunge textures, colorize black & white imagery, or create double exposures.

Step-by-Step Procedure:

1. Select your target layer in the layer stack.
2. Click the "Blend Mode" dropdown menu next to the opacity slider.
3. Choose an option (e.g., Screen for light leaks or Multiply for shadows) and observe real-time blending.

Non-Destructive Masks & Effect Layers

Functionality: Maintains an 8-bit alpha mask channel alongside layer pixels, allowing reversible cutouts. Dedicated Effect Layers isolate adjustment algorithms for modular non-destructive tuning.

When to Use: When compositing elements, smoothing subject boundaries, or experimenting with reversible color looks.

Step-by-Step Procedure:

1. Use the Eraser in mask mode or Pen path selection to carve out transparent zones on the layer.
2. Click "Invert Mask" or "Reset Mask" in the inspector to restore raw image pixels anytime.
3. Click the "Merge Visible" icon in the layer header to bake all unhidden layers into a consolidated raster.

CHAPTER 08

Professional Adjustments & Color Grading

Comprehensive adjustment engine: Exposure, RGB Tone Curves, 8-band HSL Mixer, 3-Way Wheels, and Optics.

Pixelora features an industrial-strength darkroom pipeline modeled after professional RAW developers. Divided into modular toolsets, it provides mathematical pixel control over photometric exposure, multi-channel splines, selective chromatic bands, and color balance balance.

Basic Light, Tone & White Balance

Functionality: Recalibrates dynamic range: lifts shadow detail without blowing out highlights, tunes Kelvin temperature, balances green/magenta tint, and boosts vibrance while protecting skin tones.

When to Use: As the foundational first pass for correcting underexposure, dynamic range clipping, or skewed white balance.

Step-by-Step Procedure:

1. Navigate to the Adjustments panel (press A or click the Sliders icon).
2. Tweak "Exposure" to calibrate global photometric intensity.
3. Balance "Highlights" and "Shadows" to preserve high-contrast dynamic textures.
4. Fine-tune "Temperature" and "Tint" for lifelike white balance realism.

Interactive Tone Curves (Composite & R/G/B)

Functionality: Evaluates monotonic cubic splines across an integrated histogram display. Dragging points tunes tone ranges, while switching to R/G/B channels enables cinematic split-toning cross-processing.

When to Use: To construct iconic S-curves, lift blacks for a matte film look, or color grade shadows and highlights via channel isolation.

Step-by-Step Procedure:

1. Expand the Tone Curve accordion in the Adjustments panel.
2. Click anywhere along the diagonal curve to insert a control anchor point.
3. Drag upward to brighten, downward to darken; raise the bottom-left anchor to lift blacks for a matte look.
4. Toggle R, G, and B buttons to color-cast shadows or highlights independently.

8-Channel Selective HSL Mixer

Functionality: Segregates the spectrum into 8 discrete bands (Red, Orange, Yellow, Green, Aqua, Blue, Purple, Magenta) with dedicated Hue, Saturation, and Luminance sliders per band.

When to Use: For perfecting portrait skin tones, transforming summer foliage into autumn golds, or saturating sky blues without oversaturating faces.

Step-by-Step Procedure:

1. Expand the HSL Mixer section.
2. Select between Hue, Saturation, and Luminance tabs.
3. Adjust specific sliders: for example, nudging Orange luminance upward instantly illuminates facial skin.

Cinematic 3-Way Color Grading Wheels

Functionality: Presents visual chromatic wheels targeting darks, midtones, and specular highlights, balanced by global blending and tonal distribution controls for blockbuster cinematic looks.

When to Use: To infuse photography with a deliberate, director-level color palette (e.g., Teal shadows and warm Golden Hour highlights).

Step-by-Step Procedure:

1. Open the Color Grading panel.
2. Drag the center crosshair in the Shadows wheel toward cool teal.
3. Drag the Highlights wheel crosshair toward warm amber.
4. Calibrate the "Balance" slider to govern whether highlights or shadows dominate the scene.

Detail Sharpening, Film Grain & Optics

Functionality: Sculpts micro-contrast and tactile realism: edge sharpening with masking, luminance noise reduction, clarity punch, atmospheric dehazing, vignette shading, and analog 35mm grain.

When to Use: To sharpen fine textiles and jewelry facets, eliminate high-ISO sensor noise, and frame focal subjects with elegant edge vignettes.

Step-by-Step Procedure:

1. Under the Detail tab, increase Sharpening amount to crisp up textures.
2. Under Effects, apply Clarity to pop midtone local contrast.
3. Nudge Vignette leftward to introduce a subtle, elegant focal shadow around the outer borders.

CHAPTER 09

Preset Filters & Photographic Styles

Instant stylistic photo presets across 8 categories, intensity blending slider, and before/after comparison.

Pixelora's preset library applies instant director-grade aesthetics across 8 distinct categories. Operating as true parametric transforms, each preset features an intensity blending slider (0-100%) and a split-screen Before/After modal.

The 8 Photographic Preset Families

Functionality: Presents responsive thumbnail cards across 8 families: Cinematic (high dynamic drama), Vintage (nostalgic film tones), B&W (silver halide contrast), Portrait (flattering skin softness), Warm (golden glow), and Cool (crisp icy blues).

When to Use: For rapid visual polish or establishing a uniform, branded look across social media campaigns.

Step-by-Step Procedure:

1. Press F or switch to the Filters tab in the inspector.
2. Click a category pill (e.g., Cinematic, Vintage, B&W).
3. Click any thumbnail card to apply the preset pipeline instantly to the target layer.

Filter Intensity Slider & Before/After Comparison

Functionality: The Intensity slider dynamically interpolates between default parameters and full preset values, while the Before/After modal allows side-by-side or hold-to-compare verification.

When to Use: To prevent oversaturation and evaluate whether visual enhancements genuinely elevate the photograph.

Step-by-Step Procedure:

1. Once a preset is selected, drag the "Preset Intensity" slider (e.g., to 75% for balanced subtlety).
2. Click the "Before / After" comparison icon to inspect your original and edited states side-by-side.

CHAPTER 10

Scientific Image Processing (64 Algorithms)

Point operations, spatial convolutions, edge detection, morphology, histogram CLAHE, and 2D FFT Fourier domain.

Pixelora contains an unprecedented scientific image processing engine featuring 64 algorithms executing on raw pixel buffers. Designed for computer vision engineers and academic researchers, it provides authentic mathematical implementations across spatial, morphological, histogram, and frequency domains.

Point Processing & Pixel Transforms

Functionality: Maps each input pixel value $f(x,y)$ to output $g(x,y)$ independently: luminance grayscale, negative inversion, power-law gamma curves, log transforms, and interactive threshold cutoffs.

When to Use: For machine vision pre-processing, optical document binarization, and non-linear dynamic range expansion.

Step-by-Step Procedure:

1. Select the "Point Processing" category from the filters library.
2. Pick an algorithm such as "Binary Threshold" or "Gamma Correction".
3. Use the contextual parameter slider below the intensity control to adjust the threshold or gamma coefficient in real time.

Spatial Convolutions & Noise Reduction

Functionality: Executes spatial convolution kernels over pixel neighborhoods: separable 1D Gaussian passes for 60fps performance, rank-order Median filtering for salt-and-pepper noise removal, and Bilateral filtering for bilateral edge-preserving skin smoothing.

When to Use: For cleaning high-ISO sensor noise, removing impulse speckles, and creating smooth cinematic bokeh while keeping borders razor-sharp.

Step-by-Step Procedure:

1. Select the "Blur & Smoothing" or "Noise Reduction" filter category.
2. Apply the "Bilateral Filter" or "Median Filter".
3. Adjust the radius slider to calibrate neighborhood smoothing strength.

Edge Detection Operators & Mathematical Morphology

Functionality: Extracts geometric contours and structural boundaries using gradient approximations (Sobel, Prewitt, Scharr) and Canny edge detection with hysteresis thresholding. Morphological tools perform structural erosion, dilation, morphological gradient, top-hat, and black-hat transforms.

When to Use: For computer vision feature extraction, contour vectorization, architectural blueprint line isolation, and morphological particle analysis.

Step-by-Step Procedure:

1. Select "Edge Detection" or "Morphological Operations".
2. Click "Canny Edge Detector" or "Sobel Operator".
3. Fine-tune the hysteresis thresholds to isolate prominent physical contours.

Histogram Equalization, CLAHE & 2D FFT Fourier Domain

Functionality: Features global Histogram Equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE). The 2D Cooley-Tukey Radix-2 FFT and Inverse FFT engine translates spatial data into frequency spectra, enabling Ideal/Gaussian/Butterworth Low-Pass and High-Pass filtering and periodic noise Notch suppression.

When to Use: For medical radiograph enhancement, satellite imaging, periodic noise filtering, and university-level digital signal processing demonstrations.

Step-by-Step Procedure:

1. Pick "Global / Histogram" and apply "CLAHE" to dramatically elevate local contrast without blowing out highlights.
2. Choose "Frequency Domain" to apply genuine 2D FFT Butterworth or Gaussian frequency filters.

CHAPTER 11

Background Removal & Content-Aware Inpainting

Automatic background cutout, manual edge refinement, 40+ background templates, and Navier-Stokes inpainting.

Pixelora integrates automatic background segmentation alongside content-aware texture synthesis. Users can strip backdrops in 1 click, swap in pre-rendered backdrops, or paint over unwanted elements and watermarks using the Navier-Stokes inpainting engine.

Automatic & Manual Background Cutout

Functionality: Segments foreground subjects into pristine alpha channels. Includes manual restore and erase brushes to perfect difficult contours such as hair strands and transparent glassware.

When to Use: For e-commerce product listings, promotional sticker cutouts, and creative composite collages.

Step-by-Step Procedure:

1. Press R or select "Remove Background" from the toolbox.
2. Click the primary "Auto Remove Background" button to execute automatic segmentation.
3. Refine intricate borders using the manual mask brush or eraser tools.

Content-Aware Object Removal & Inpainting (J)

Functionality: Allows painting over distracting objects with a mask brush; the client-side inpainting engine propagates boundary gradients and texture into the hole via Telea / Navier-Stokes equations.

When to Use: To erase photo-bombers, power lines, scratch defects on physical products, watermarks, or dated timestamps.

Step-by-Step Procedure:

1. Press J to switch into Remove Object mode.
2. Paint over the unwanted item using the red highlight mask brush.
3. Click "Apply Removal" to trigger the inpainting solver and reconstruct the background.

Background Templates Library & Custom Uploads

Functionality: Applies full-bleed backgrounds beneath isolated product layers, with instant support for custom backdrop file uploads.

When to Use: To transform mundane tabletop phone snapshots into high-end commercial ad campaigns.

Step-by-Step Procedure:

1. Press G or switch to the Backgrounds tab in the inspector.
2. Explore categories: Studio Minim, Luxury Marble, Warm Wood, or Nature.
3. Click any backdrop to inject it directly beneath your foreground cutout layer.

CHAPTER 12

High-Resolution Export Engine

Production formats: PNG, JPG, WEBP, SVG, compression ratios, scaling factors, and alpha transparency.

Pixelora's export pipeline renders pristine raster composites directly from canvas layers. It supports exact pixel scaling (1x, 2x, 3x, 4x), fine-tuned JPEG/WEBP compression quality, PNG alpha transparency preservation, and SVG vector pass-through.

Format Matrix: PNG vs JPG vs WEBP vs SVG

Functionality: PNG provides lossless raster quality and alpha transparency; JPG offers compact file sizes for photos; WEBP delivers next-gen compression with transparency; SVG preserves vector scalability.

When to Use: Use PNG for logos and cutouts with transparent backdrops; JPG for social photography; WEBP for high-performance e-commerce pages.

Step-by-Step Procedure:

1. Press Ctrl+E or click the top "Export" button.
2. Select your target format: PNG, JPG, WEBP, or SVG.
3. For JPG/WEBP, set the Quality slider between 80% and 100%.
4. Click "Download Image" to trigger immediate browser file saving.

Parameter	Type	Default	Description
Scale Factor	1x, 2x, 3x	1x	Multiplies canvas resolution for crisp presentation on ultra-high-DPI Retina screens.
Preserve Transparency	Boolean	True (PNG/WEBP)	Ensures canvas regions without background fills remain completely transparent.

CHAPTER 13

Master Keyboard Shortcuts Reference

Complete organized table of all verified shortcuts for tools, layers, view, and document commands.

Pixelora’s keyboard shortcuts follow industry conventions, empowering designers to execute high-speed workflows without lifting their hands from the keyboard.

Command / Action	Key	Category & Scope
New Project	Ctrl + N	Open new project dialog to set canvas dimensions [General & Project]
Open Project	Ctrl + O	Open a previously saved project from local browser storage [General & Project]
Save Project	Ctrl + S	Save current project state to browser local storage [General & Project]
Export Image	Ctrl + E	Open export dialog with PNG, JPG, WebP, SVG formats and up to 3x resolution scaling [General & Project]
Undo	Ctrl + Z	Undo the last action performed on the canvas [General & Project]
Redo	Ctrl + Y / Shift + Ctrl + Z	Redo the previously undone action [General & Project]
Keyboard Shortcuts Help	? / F1 / Ctrl + /	Open the interactive keyboard shortcuts cheat sheet modal [General & Project]
Select / Move Tool	V	Select, drag, and position layers on the canvas stage [Tools]
Advanced Transform Tool	Ctrl + T / K	Activate advanced transformation handles for position, dimensions, rotation, and skew [Tools]
Crop Tool	C	Crop canvas frame with aspect ratio presets or custom freeform and circular bounding [Tools]
Hand / Pan Tool	H	Pan and navigate across the canvas viewport without moving layers [Tools]
Zoom Tool	Z	Activate zoom magnification tool to zoom into canvas details [Tools]
Brush Tool	B	Draw freehand raster strokes with size, opacity, and hardness parameters [Tools]
Pen Tool	P	Draw vector paths and Bézier curves with editable anchor nodes and stroke controls [Tools]
Eraser Tool	E	Erase portions of drawing layers or layer masks with interactive circular bounds [Tools]
Add Text Tool	T	Insert customizable typography layers with Arabic/Latin fonts and text styles [Tools]
Shapes & Icons Library	U	Open shape library dialog to insert rectangles, circles, stars, and polygons [Tools]
Remove Object Tool	J	Paint over unwanted elements to seamlessly inpaint and reconstruct

		backdrop texture [Tools]
Preset Filters Panel	F	Open preset filters tab to apply any of the 64 mathematical or cinematic filters [Tools]
Adjustments & Color Panel	A	Open adjustments panel for Exposure, Curves, HSL Mixer, and 3-Way Color Wheels [Tools]
Remove Background Tool	R	Open background removal tab with 1-click auto cutout and edge refinement brushes [Tools]
Background Library Tab	G	Open backgrounds library tab for solid colors, gradients, textures, and studio scenes [Tools]
Cut Layer	Ctrl + X	Copy active layer to clipboard and remove it from canvas [Layers & Editing]
Copy Layer	Ctrl + C	Copy active layer data into internal memory [Layers & Editing]
Paste Layer	Ctrl + V	Paste previously copied layer with slight offset [Layers & Editing]
Duplicate Layer	Ctrl + D / Ctrl + J	Duplicate selected layer immediately into a fresh top layer [Layers & Editing]
Delete Layer	Delete / Backspace	Delete currently selected layer from project layers [Layers & Editing]
Select Layer / All	Ctrl + A	Select active canvas element [Layers & Editing]
Deselect / Cancel	Escape	Deselect active layer, dismiss modals, or cancel current operation [Layers & Editing]
Bring Forward	Ctrl +]	Move active layer one step up in the layer hierarchy stack [Layers & Editing]
Bring to Front	Shift + Ctrl +]	Move active layer to the absolute top of the layer hierarchy stack [Layers & Editing]
Send Backward	Ctrl + [	Move active layer one step down in the layer hierarchy stack [Layers & Editing]
Send to Back	Shift + Ctrl + [	Move active layer to the bottom of the layer stack above canvas background [Layers & Editing]
Decrease Brush Size	[	Decrease active brush or eraser stroke diameter by 5px [Layers & Editing]
Increase Brush Size	]	Increase active brush or eraser stroke diameter by 5px [Layers & Editing]
Nudge Layer (1px)	⬅️ ⬆️ ⬇️ ⬅️ ⬆️ ⬇️ ⬅️ ⬆️ ⬇️	Nudge selected layer by 1px (Hold Shift for 10px leap) [Layers & Editing]
Image Size (Resample)	Alt + I	Resize image dimensions with resampling

		algorithms and DPI adjustments [Layers & Editing]
Canvas Size (9-point Anchor)	Alt + C	Expand or trim canvas frame margins using 9-point anchor orientation [Layers & Editing]
Zoom In	Ctrl + + / Ctrl + =	Increase canvas magnification factor by 15% [View & Navigation]
Zoom Out	Ctrl + - / Ctrl + _	Decrease canvas magnification factor by 15% [View & Navigation]
Fit to Screen & Center	Ctrl + 0	Scale canvas to fit visible viewport and reset pan offset to center [View & Navigation]
Actual Size (100%)	Ctrl + 1	Set zoom scale to 100% pixel-to-pixel ratio [View & Navigation]

CHAPTER 14

Troubleshooting & Diagnostics

Practical recovery procedures for canvas responsiveness, image ingestion, alpha transparency, and fonts.

A structured diagnostic guide offering verified causes and step-by-step remedies for common editing anomalies.

Anomaly	Probable Root Cause	Verified Resolution
Brush or Tool not responding on the canvas	Target layer is locked, hidden, or is a vector/text layer that does not accept direct raster drawing.	Select an active raster layer or click "+" to add a new drawing layer. Ensure the layer is unlocked and visible.
Exported image has a solid white background instead of transparency	File was exported as JPG (which lacks alpha channels) or canvas background fill was set to solid white.	In the Export modal, choose PNG or WEBP, check "Preserve Transparency", and verify the canvas background is set to transparent.
Cannot select or move a specific element on the stage	An upper layer with large bounding bounds is intercepting pointer clicks above the desired element.	Click the specific layer card directly in the right-hand Layers panel to force active selection.
Noticeable lag when applying complex mathematical filters	Image dimensions exceed ultra-high thresholds (e.g. >16MP), requiring millions of convolution matrix calculations.	Open the Image Size dialog and resample dimensions to a manageable web resolution (e.g., 2000px) before heavy convolutions.
Arabic text glyphs appear disconnected or reversed in legacy browsers	Legacy browser environment missing native Unicode bidirectional OpenType Arabic shaping.	Update your browser to the latest version of Chrome, Edge, Safari, or Firefox to leverage standard BiDi engines.
Auto background cutout trimmed away subtle parts of the product	Low color contrast between the subject and backdrop (e.g. white sneaker on white studio wall).	Use the "Restore Mask Brush" in the Background Removal panel to hand-paint and recover the omitted edges.

CHAPTER 15

Frequently Asked Questions (FAQ)

Authoritative answers to core user queries regarding privacy, storage, limitations, and features.

Curated compilation of core questions frequently posed by creative professionals and student researchers.

Q: Does Pixelora upload my personal photos to any remote servers?

Never. Pixelora is architected as a local-first workstation. All pixel computations, convolutions, and layer compositions execute in your browser's local client memory, ensuring absolute privacy.

Q: What is the difference between "Image Size" and "Canvas Size"?

"Image Size" resamples and re-interpolates the document's pixel grid to scale contents. "Canvas Size" expands or shrinks the outer frame boundary (adding borders or clipping margins) without resizing individual layers.

Q: What is the difference between Filters and Adjustments?

Adjustments are parametric continuous controls for specific photometric values (Exposure, Tone Curves, HSL). Filters are holistic recipes and mathematical algorithms (Cinematic looks, FFT, Edge Detectors) applied as a single composite transformation.

Q: What is Non-Destructive Editing?

It ensures your raw source imagery remains pristine. You can re-adjust filters, reshape masks, rewrite text, or modify opacity at any future point without generational quality degradation.

Q: How do I export my artwork at peak resolution for large-format printing?

In the Export modal (Ctrl+E), select PNG or WEBP, increase the Scale Factor to 2x or 3x, and verify quality is set to maximum for multi-megapixel print readiness.

Q: Can I use Pixelora on tablets and mobile devices?

Yes, Pixelora's responsive layout adapts seamlessly to tablets and touch devices, supporting stylus input, pinch-to-zoom, and touch dragging.